Sahaj Bhadja

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Summary

Software Engineer with an MS in CS from USC and 2+ years building data infrastructure in **Python**: SQL query and index optimization, **Spark** and ETL pipelines, and concurrent job orchestration, with **C++** systems work and DBMS coursework.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tuned access patterns and indexes for multi-tenant Azure SQL and PostgreSQL schemas with row-level security, cutting dashboard latency from 1.2s to 350ms for 500+ users through **query optimization**.
- Built a fault-tolerant async job system with explicit state transitions and failure isolation, plus SQL transformation pipelines across PostgreSQL and SQL Server, tripling throughput and sustaining 99%+ **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed a **concurrent** batch-processing engine with Python ThreadPoolExecutor for PDF, DOCX, and ZIP ingestion, processing 1K+ files per batch with fault-tolerant execution behind 10K+ daily API requests.
- Built a **Python** backend with model routing and timeout fallback across GPT-4o, Claude, and Qwen-VL, persisting per-call usage in normalized MySQL tables and cutting feedback turnaround 60% for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built end-to-end **data pipelines** processing 100K+ trajectory logs and HD-map vectors into 10Hz time-aligned features for an offline-RL GPT-based Decision Transformer, improving long-horizon planning 18% in simulation.
- Designed a spatial-temporal attention encoder in PyTorch with reproducible Hydra configs and W&B tracking, benchmarking it against three baselines in closed-loop **simulation** and improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** on AWS Glue and S3 that transformed raw JSON, CSV, and Parquet into normalized formats across raw, intermediate, and final storage tiers, boosting pipeline efficiency 68%.
- Optimized large-scale API integrations with indexing, **caching**, and pagination, improving response times 30%, and automated nightly cross-platform builds and 20+ test suites on Windows, Linux, and macOS.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built hourly **Spark** micro-batch pipelines on Databricks over 50K+ booking and payment events, aggregating peak hours, service demand, daily active users, and revenue for internal dashboards and client reporting.
- Developed Python Flask APIs with Firebase real-time sync and **transactional locking** against double-booked slots, sustaining sub-200ms latency and adding fraud-checked, idempotent payment webhooks that cut manual handling 70%.

Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, Shell
Systems & Databases: Query Optimization, Indexing, Database Systems, Operating Systems, Concurrency, Distributed Systems, Scheduling Algorithms
Data Infrastructure: Apache Spark, Databricks, AWS Glue, ETL Pipelines, Incremental Loads, PostgreSQL, SQL Server, Snowflake
ML: PyTorch, Transformers, Feature Engineering, Hydra, W&B
Backend & Infra: FastAPI, Flask, REST APIs, Docker, Kubernetes, AWS, Linux, Git, CI/CD
Practices: Data Structures, Algorithms, Unit Testing, PyTest, Debugging

Projects

Python Feature Query Planner | *Python, C++, AST, pybind11, PostgreSQL*

- Designed a feature-pipeline **query planner** that parses Python feature definitions via AST into a dependency DAG, applies filter pushdown and lookup batching, and executes plans on a vectorized **C++** engine.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** with C++ and a React simulator, testing 1K+ process combinations and demonstrating up to 25% lower average turnaround.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Snowflake, SQLite*

- Extended a Java and TypeScript database workspace spanning 10+ engines including PostgreSQL, Snowflake, and SQLite, generating schema-aware SQL with **validated execution** at 82% accuracy across relational databases.